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UNIT- 03 (MATHS- 04)

Qu-01 Calculate the First Four Central moment about mean of Following data.

x	0	1	2	3	4	5	6	7	8
f	1	8	28	56	70	56	28	8	1

[AKTU - 2021-22]

x	f	$x \times f$	$x-\bar{x}$	$f(x-\bar{x})$	$f(x-\bar{x})^2$	$f(x-\bar{x})^3$	$f(x-\bar{x})^4$
0	1	0	-4.21	-4.21			
1	8	8	-3.21	-25.68			
2	28	56	-2.21	-61.88			
3	56	168	-1.21	-67.76			
4	70	280	-0.21	-14.7			
5	56	280 ~~313.6~~	~~0.78~~	~~43.68~~			
6	28	168	1.79	50.12			
7	8	56	2.79	22.32			
8	1	8	3.79	3.79			

$\Sigma f = 256 \quad \Sigma fx = 1080 \quad \Sigma f(x-\bar{x}) = -54.32$

$\bar{x} = \frac{\Sigma fx}{N} = \frac{1080}{256} = 4.21 \quad \boxed{\bar{x} = 4.21}$

$\mu_1 = \frac{f(x-\bar{x})^1}{N} = \frac{-54.32}{256} = -0.21 \quad \boxed{\mu_1 = -0.21}$

$\mu_2 = \frac{f(x-\bar{x})^2}{N}$

$\mu = \frac{f(x-\bar{x})^{\cancel{1}\,2}}{N}$

$\mu_3 = \frac{f(x-\bar{x})^3}{N}$

$\mu_4 = \frac{\Sigma f(x-\bar{x})^4}{N}$

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TRICK :-

$1 \quad \mu_1' \quad \mu_1'^2 \quad \mu_1'^3$
μ_1^0

$\mu_1 = 0$

$\mu_2 = \mu_2' - \mu_1'$

$\mu_3 = \mu_3' - \mu_2' + \mu_1'$

$\mu_2 = \mu_2' - \mu_1'^2$

$\mu_3 = \mu_3' - 3\mu_1'\mu_2' + 2\mu_1'^3 \quad [1 \quad 3 \quad 2]$

$\mu_4 = \mu_4' - 4\mu_1'\mu_3' + 6\mu_1'^2\mu_2' - 3\mu_1'^4 \quad [1 \quad 4 \quad 6 \quad 3]$

Qu-02 The First Four moment of a distribution about the Value of 4 the Variable are -1.5, 17, -30 and 80. Find the moment μ_1', μ_2, μ_3, μ_4 and also β_1 and β_2? [AKTU - 2022-23]

Sol^n $A = 4$

$\mu_1' = -1.5$, $\mu_2' = 17$, $\mu_3' = -30$, $\mu_4' = 80$

Moment about mean

$\boxed{\mu_1 = 0}$

$\mu_2 = \mu_2' - \mu_1'^2$

$= 17 - (-1.5)^2$

$\boxed{\mu_2 = 14.75} = 14.75$

$\mu_3 = \mu_3' - 3\mu_1'\mu_2' + 2\mu_1'^3$

$= -30 - 3 \times -1.5 \times 17 + 2(-1.5)^3$

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$\boxed{\mu_3 = 39.7}$

$\mu_4 = \mu_4' - 4\mu_1'\mu_3' + 6\mu_1'^2\mu_2' - 3\mu_1'^4$

$= 80 - 4 \times -1.5 \times 30 + 6 \times (-1.5)^2 \times 17 - 3 \times (-1.5)^4$

$\boxed{\mu_4 = 114.312}$

Karl Pearson Coefficients.

- $\beta_1 = \frac{\mu_3^2}{\mu_2^2} = \frac{(39.7)^2}{(14.75)^2} = \frac{1576.09}{217.5625} = 7.2443$

- $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{114.312}{(14.75)^2} = \frac{114.312}{217.5625} = 0.5254$ Ans

Qu-03 Using method of least Square Find the Curve $y = ab^x$ that fit the data

x	2	3	4	5	6
y	8.3	15.4	33.1	65.2	127.4

[AKTU-2020-21]

OR

Fit the Parabolla Curve of Second degree to the Following data

X	0	1	2	3	4
Y	1	1.8	1.3	2.6	6.3

[AKTU - 2022-23]

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$y = ab^x$ taking log on both side

$\log y = \log ab^x$

$\log y = \log a + \log b^x$ $\qquad \log m^n = n \log m$

$\log y = \log a + x \log b$

$Y = A + B X$

$Y = \log y$, $A = \log a$, $x = \log b$

$\Sigma Y = An + B \Sigma x$ $\qquad A\Sigma n$

$\Sigma xY = A\Sigma x + B\Sigma x^2$ [multiply by x]

Parabola:-

$y = a + bx + cx^2$

$\Sigma y = an + b\Sigma x + c\Sigma x^2$ — (1)

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$ — (2)

$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$ — (3)

X	Y	x^2	x^3	x^4	xy	x^2y
0	1	0	0	0	0	0
1	1.8	1	1	1	1.8	1.8
2	1.3	4	8	16	2.6	5.2
3	2.5	9	27	81	7.5	22.5
4	6.3	16	64	256	25.2	100.8
10	12.9	30	100	354	37.1	130.3

$\Sigma y = an + b\Sigma x + c\Sigma x^2$

$12.9 = 5a + 10b + 30c$ —(1)

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$

$37.1 = 10a + 30b + 100c$ —(2)

$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$

$130.3 = 30a + 100b + 354c$ —(3)

a =	b =	c =

$$\boxed{y = a + bx + cx^2}$$

Qu-4 Fit a Parabolic Curve of Regression y on x to the Following data

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	1.1	1.3	1.6	2.0	2.7	3.4	4.1

[AKTU - 2021-22]

x	y	xy	x2
1.0	1.1	1.1	1
1.5	1.3	1.95	2.25
2.0	1.6	3.2	4
2.5	2.0	5	6.25
3.0	2.7	8.1	9
3.5	3.4	11.9	12.25
4.0	4.1	16.4	16
17.5	16.2	47.65	50.75

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$$y \text{ on } x = b_{yx} = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$$b_{yx} = \frac{7 \times 47.65 - 17.5 \times 16.2}{7 \times 50.75 - (17.5)^2}$$

$$= \frac{333.55 - 283.5}{355.25 - 306.25}$$

$$= \frac{50.05}{49} \qquad \boxed{b_{yx} = 1.021}$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{17.5}{7} \qquad \boxed{\bar{x} = 2.5}$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{16.2}{7} \qquad \boxed{\cancel{\bar{y} = 3.24}} \quad \boxed{\bar{y} = 2.3142}$$

$$(y - \bar{y}) = b_{yx}(x - \bar{x})$$

$$\boxed{(y - \overset{2.3142}{\cancel{3.24}}) = 1.021(x - 2.5)} \text{ Ans}$$

Qu-05 The First Four moment of a distn about $x = 4$ are 1, 4, 10 and 45, Obtain the Various Characterstics of the distribution on the basis of the given infm comment Upon the nature of the distribution:

[AKTU - 2016, 2018]

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Sol^n $A = 4$, $\mu_1' = 1$, $\mu_2' = 4$, $\mu_3' = 10$, $\mu_4' = 45$

$\mu_1 = 0$

$\mu_2 = \mu_2' - \mu_1'^2 = 4 - 1^2 = 3$ $\boxed{\mu_2 = 3}$

$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$
$= 10 - 3 \times 4 \times 1 + 2 \times 1^3$
$\boxed{\mu_3 = 0}$

$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$
$= 45 - 4 \times 10 \times 1 + 6 \times 4 \times 1 - 3 \times 1$
$\boxed{\mu_4 = 26}$

Kurtosis: $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{26}{9} = 2.89 < 3$

$\boxed{\beta_2 < 3}$ Platykurtic Ans

Qu-06 The Following data regarding the height (y) and the weight (x) of 100 College Student is given

$\Sigma x = 15000$ $\Sigma x^2 = 2272500$
$\Sigma y = 6800$ $\Sigma y^2 = 463025$
$\Sigma xy = 1022250$

find you x ?

[AKTU - 2012, 2017]

Sol^n $b_{yon x} = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$

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$$b_{yx} = \frac{100 \times 1022250 - 15000 \times 6800}{100 \times 2272500 - (15000)^2}$$

$$\boxed{b_{yx} = 0.1}$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{15000}{100} = \boxed{\bar{x} = 150}$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{6800}{100} = \boxed{\bar{y} = 68}$$

$$(y - \bar{y}) = b_{yx}(x - \bar{x})$$
$$(y - 68) = 0.1(x - 150)$$

$$\boxed{y = 0.1x + 53}$$ Ans

MOMENT Generating Function :-

$$\boxed{M_x(t) = \Sigma e^{tx} f(x)}$$

Qu-07 Find the moment generating Fun^n of the discrete binomial dist^n given by $P(x) = {}^nC_x p^x q^{n-x}$ (where $q = 1-p$) Also find the first and second moments about the mean.

[AKTU-2020-21]

Sol^n $m_x(t) = \Sigma e^{tx} P(x)$

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$M_x(t) = \Sigma e^{tx} p(x)$

$= \Sigma e^{tx} \, {}^nC_x \, p^x q^{n-x}$ $\qquad e^{tx} \cdot p^x$

$(e^t p)^x$

$= \sum_{x=0}^{n} {}^nC_x \, (pe^t)^x q^{n-x}$

$\boxed{M_x(t) = (q + pe^t)^n}$ $\qquad (a+z)^n = \sum_{x=0}^{n} {}^nC_x \, z^x a^{n-x}$

$\bar{x} = \text{mean} = v_1 = \left[\frac{d}{dt} m_x(t)\right]_{t=0}$ $\qquad x^n = nx^{n-1}$

$= [n(q+pe^t)^{n-1} \times pe^t]_{t=0}$ $\qquad e^t = e^t$

$e^0 = 1$

$= [n(q+p)^{n-1} \cdot p]$ $\qquad \therefore p+q=1$

$q = 1-p$

$\text{mean} = \bar{x} = np = v_1$ $\qquad p+q=1$

$v_2 = \left[\frac{d^2}{dt} m_x(t)\right]_{t=0}$

$= np\,[e^t(n-1)(q+pe^t)^{n-2} pe^t + (q+pe^t)^{n-1} e^t]_{t=0}$

$= np\,[(n-1)(q+p)^{n-2} p + (q+p)^{n-1}]$ $\qquad \therefore p+q=1$

$= np\,[(n-1)p + 1]$

$= np\,[np - p + 1]$ $\qquad q = 1-p \qquad p+q=1$

$= np\,[np + q]$

$\boxed{v_2 = n^2p^2 + npq}$

$\boxed{\mu_1 = 0}$

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$\mu_2 = \nu_2 - \bar{x}^2$

$n^2p + npq - n^2p^2$

$\boxed{\mu_2 = npq}$

Qu-08 ~~The~~ If the regression cofficient is 0.8 and 0.2. what will be the coffient of correlation? [AKTU-2020-21]

Sol^n $r_x = 0.8$

$r_y = 0.2$

Cofficient $= \sqrt{r_x r_y}$

$= \sqrt{0.8 \times 0.2} = \sqrt{0.16} = 0.4$

$\boxed{\text{Coffient of Correlation is } 0.4}$ Ans

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